

CHE 301 Assignment #01

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Abstract

This assignment integrates classical thermodynamics with numerical methods to reinforce both analytical reasoning and computational problem-solving skills central to chemical engineering. In Problem A, the ideal Brayton cycle is analyzed under steady-state, ideal-gas assumptions with constant heat capacity and isentropic compression and expansion. Given fixed inlet conditions and a maximum turbine inlet temperature, the cycle performance is evaluated by deriving and applying thermodynamic relationships for temperature, pressure, work, and efficiency. The problem emphasizes the physical interpretation of compressor pressure ratio and its role in optimizing net power output, while also reinforcing the distinction between maximum power operation and maximum efficiency operation.

A key objective of the Brayton cycle analysis is to determine the optimal compressor pressure ratio using both analytical calculus-based methods and numerical optimization via Excel Solver. Once the optimal condition is identified, the assignment requires calculation of state variables throughout the cycle, including compressor discharge conditions and turbine exhaust temperature, followed by evaluation of thermal efficiency. The Brayton cycle efficiency is then compared against the theoretical Carnot efficiency operating between the same temperature limits, providing insight into fundamental thermodynamic limits versus practical cycle performance. Finally, the cycle analysis is extended to an engineering design context by calculating the minimum air flow rate required to produce a specified power output, expressed in MMSCFD.

Problem B shifts focus to numerical root-finding techniques applied to a nonlinear algebraic equation describing vapor–liquid equilibrium in a multicomponent hydrocarbon mixture. Students derive the analytical derivative of the vapor fraction equation and implement both the Newton and Secant methods to compute the vapor fraction starting from an initial guess. Iterative convergence behavior is documented in tabular form and verified using Excel’s Goal Seek function, reinforcing numerical accuracy and validation practices. Together, the two problems bridge thermodynamics, numerical analysis, and spreadsheet-based computation, highlighting how analytical models and numerical tools are jointly used to solve realistic chemical engineering problems.

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1 Problem A: Ideal Brayton Cycle Analysis

1.1 Background

The Brayton cycle is a fundamental thermodynamic model used to describe the operation of gas turbine engines in applications such as power generation and aircraft propulsion. It consists of four idealized processes: isentropic compression of air in a compressor, constant-pressure heat addition in a combustor, isentropic expansion through a turbine, and constant-pressure heat rejection to the surroundings. Under ideal gas and steady-state assumptions, the temperatures and pressures at each state are related through isentropic relations that depend on the ratio of specific heats, allowing cycle performance to be expressed analytically in terms of the compressor pressure ratio. Key performance metrics of the Brayton cycle include net power output, thermal efficiency, and work balance between the compressor and turbine. Understanding how these quantities vary with pressure ratio and temperature limits is essential for designing efficient gas turbine systems and for distinguishing between operating conditions that maximize power output versus those that maximize thermal efficiency.

1.2 Problem 1a: Compressor Pressure Ratio for Maximum Net Power Output

The compressor pressure ratio that maximizes the net power output of an ideal Brayton cycle is defined as

$$r_p \equiv \frac{P_2}{P_1}.$$

For an ideal Brayton cycle with constant C_p and isentropic compressor and turbine,

$$w_{\text{net}} = w_{\text{turbine}} - w_{\text{compressor}}.$$

For a steady-flow device, such as a compressor or turbine,

$$\dot{Q} - \dot{W} = \dot{m}(h_{\text{out}} - h_{\text{in}}).$$

The process is adiabatic, so $\dot{Q} = 0$. This leaves

$$-\dot{W} = \dot{m}(h_{\text{out}} - h_{\text{in}}).$$

The molar flow rate is equal through all components, so writing the equation per mole using $W = \dot{m}w$ gives

$$-w = h_{\text{out}} - h_{\text{in}}.$$

This can be written as

$$w = h_{\text{in}} - h_{\text{out}}.$$

For the compressor, the inlet is state 1 and the outlet is state 2:

$$w_c = h_1 - h_2.$$

Using $h = C_p T$, this becomes

$$w_c = C_p(T_1 - T_2).$$

The magnitude of the compressor work is therefore

$$w_c = C_p(T_2 - T_1).$$

For the turbine, the inlet is state 3 and the outlet is state 4:

$$w_t = C_p(T_3 - T_4).$$

The net work is

$$w_{\text{net}} = C_p [(T_3 - T_4) - (T_2 - T_1)].$$

This expression is the exact net work of the cycle per mole. Since C_p is a given positive constant, maximizing w_{net} is equivalent to maximizing

$$f = (T_3 - T_4) - (T_2 - T_1),$$

such that

$$w_{\text{net}} = C_p f.$$

For an isentropic temperature–pressure relation,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{(\gamma-1)/\gamma}.$$

Defining

$$k = \frac{\gamma - 1}{\gamma},$$

and using $r_p = P_2/P_1$, this becomes

$$\frac{T_2}{T_1} = r_p^k, \quad T_2 = T_1 r_p^k.$$

For the turbine,

$$\frac{T_4}{T_3} = \left(\frac{P_4}{P_3} \right)^k.$$

The turbine inlet pressure is $P_3 = P_2$ and the turbine outlet pressure is $P_4 = P_1$, since the Brayton cycle is a closed pressure loop. Therefore,

$$\frac{T_4}{T_3} = \left(\frac{P_1}{P_2} \right)^k = \left(\frac{P_2}{P_1} \right)^{-k}, \quad T_4 = T_3 r_p^{-k}.$$

Substituting the isentropic relations into the net work expression gives

$$f(r_p) = [T_3 - T_3 r_p^{-k}] - [T_1 r_p^k - T_1].$$

This simplifies to

$$f(r_p) = T_3 (1 - r_p^{-k}) - T_1 (r_p^k - 1) .$$

Differentiating with respect to r_p ,

$$\frac{df}{dr_p} = T_3 (-(-kr_p^{-k-1})) - T_1 (kr_p^{k-1}) .$$

This becomes

$$\frac{df}{dr_p} = T_3 kr_p^{-k-1} - T_1 kr_p^{k-1} .$$

Factoring out kr_p^{-1} ,

$$\frac{df}{dr_p} = kr_p^{-1} [T_3 r_p^{-k} - T_1 r_p^k] .$$

Setting the derivative equal to zero,

$$0 = T_3 r_p^{-k} - T_1 r_p^k .$$

This gives

$$T_3 r_p^{-k} = T_1 r_p^k, \quad T_3 = T_1 r_p^{2k} .$$

Solving for r_p ,

$$r_p^{2k} = \frac{T_3}{T_1}, \quad r_p = \left(\frac{T_3}{T_1} \right)^{1/(2k)} .$$

Since $k = (\gamma - 1)/\gamma$,

$$r_p = \left(\frac{T_3}{T_1} \right)^{\gamma/[2(\gamma-1)]} .$$

Substituting numerical values,

$$r_p = \left(\frac{1400 \text{ K}}{300 \text{ K}} \right)^{1/[2(1.4-1)/1.4]} = 14.82 .$$

1.3 Problem 1b

Completed in Excel.

	A	B	C	D	E
1	T1	300		T2	T4
2	T3	1400		648.0741	648.0741
3	Gamma	1.4			
4	Cp	29.1		wnet	11752.09
5	Exponent k	0.285714			
6					
7	rp	14.81705			

Figure 1: Solver used to determine the compressor pressure ratio that maximizes the net power output of the cycle

1.4 Problem 2: State Conditions at Optimal Pressure Ratio

Since $r_p = P_{\text{out}}/P_{\text{in}}$ for the compressor,

$$r_p = \frac{P_2}{P_1} = 14.82.$$

Solving for P_2 ,

$$P_2 = r_p P_1 = (14.82)(100 \text{ kPa}) = 1482 \text{ kPa}.$$

Using the isentropic compression relation,

$$\frac{T_2}{T_1} = r_p^k, \quad k = \frac{1.4 - 1}{1.4} = 0.2857,$$

$$T_2 = (300 \text{ K})(14.82^{0.2857}) = 648.09 \text{ K}.$$

For the turbine,

$$T_4 = T_3 r_p^{-k} = (1400 \text{ K})(14.82^{-0.2857}) = 648.06 \text{ K}.$$

Thus,

$$T_4 = T_2.$$

1.5 Problem 3: Thermal Efficiency of the Brayton Cycle at Maximum Net Power

For a heat engine, the thermal efficiency is defined as

$$\eta_{\text{th}} = \frac{W_{\text{net}}}{Q_{\text{in}}}.$$

The net work is

$$W_{\text{net}} = W_{\text{turbine}} - W_{\text{compressor}}.$$

Heat added in the combustor cannot come from the compressor or turbine because they are adiabatic.

For the compressor,

$$\dot{Q} - \dot{W} = \dot{n}(h_{\text{out}} - h_{\text{in}}),$$

so

$$-W_{\text{compressor}} = h_2 - h_1.$$

Since $h = C_p T$, the compressor work per mole is

$$-W_{\text{compressor}} = C_p(T_2 - T_1).$$

Because work is done on the fluid in the compressor, the signed compressor work is negative. Defining the compressor work input as

$$W_{\text{compressor,in}} \equiv -W_{\text{compressor}},$$

we obtain

$$W_{\text{compressor,in}} = C_p(T_2 - T_1).$$

For the turbine,

$$\dot{Q} - \dot{W} = \dot{n}(h_{\text{out}} - h_{\text{in}}),$$

so

$$-W_{\text{turbine}} = h_4 - h_3.$$

Since $h = C_p T$, the turbine work per mole is

$$-W_{\text{turbine}} = C_p(T_4 - T_3).$$

Thus,

$$W_{\text{turbine}} = -C_p(T_4 - T_3) = C_p(T_3 - T_4).$$

The net work is therefore

$$W_{\text{net}} = C_p(T_3 - T_4) - C_p(T_2 - T_1),$$

which can be written as

$$W_{\text{net}} = C_p [(T_3 - T_4) - (T_2 - T_1)].$$

Since $T_2 = T_4$, this becomes

$$W_{\text{net}} = C_p(T_3 - 2T_2 + T_1).$$

For the combustor,

$$\dot{Q} - \dot{W} = \dot{n}(h_{\text{out}} - h_{\text{in}}),$$

with no work done, so

$$Q_{\text{in}} = h_3 - h_2.$$

Since $h = C_p T$,

$$Q_{\text{in}} = C_p(T_3 - T_2).$$

Thus, the thermal efficiency is

$$\eta_{\text{th}} = \frac{C_p(T_3 - 2T_2 + T_1)}{C_p(T_3 - T_2)}.$$

Canceling C_p ,

$$\eta_{\text{th}} = \frac{T_3 - 2T_2 + T_1}{T_3 - T_2}.$$

Substituting numerical values,

$$\eta_{\text{th}} = \frac{1400 - 2(648) + 300}{1400 - 648} = 0.537 = 53.7\%.$$

1.6 Problem 4: Carnot Efficiency Comparison

The Carnot efficiency is given by

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}.$$

The cold reservoir temperature is

$$T_C = 300 \text{ K},$$

and the hot reservoir temperature is

$$T_H = 1400 \text{ K}.$$

Thus,

$$\eta_{\text{Carnot}} = 1 - \frac{300}{1400} = 0.7857 = 78.6\%.$$

1.7 Problem 5: Maximum Thermal Efficiency Condition

For an ideal Brayton cycle with fixed temperature limits, maximum thermal efficiency occurs when the compressor exit temperature equals the turbine inlet temperature, so

$$T_2 = T_3.$$

For a heat engine,

$$\eta_{\text{th}} = \frac{W_{\text{net}}}{Q_{\text{in}}}.$$

The net work is

$$W_{\text{net}} = W_{\text{turbine}} - W_{\text{compressor}}.$$

Using the same expressions for compressor and turbine work,

$$W_{\text{net}} = C_p(T_3 - T_4) - C_p(T_2 - T_1).$$

When $T_2 = T_3$,

$$W_{\text{net}} = C_p [(T_2 - T_4) - (T_2 - T_1)] = C_p(T_1 - T_4).$$

For an isentropic process,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^k, \quad k = \frac{\gamma - 1}{\gamma}.$$

Since $T_2 = T_3$,

$$\frac{T_3}{T_1} = r_p^k.$$

For the turbine,

$$\frac{T_4}{T_3} = \left(\frac{P_4}{P_3}\right)^k.$$

With $P_3 = P_2$ and $P_4 = P_1$,

$$\frac{T_4}{T_3} = \left(\frac{P_2}{P_1}\right)^{-k} = r_p^{-k}.$$

Since

$$\frac{T_3}{T_1} = r_p^k,$$

it follows that

$$\frac{T_1}{T_3} = r_p^{-k}.$$

Setting

$$\frac{T_1}{T_3} = \frac{T_4}{T_3},$$

the T_3 terms cancel, giving

$$T_1 = T_4.$$

Substituting into the net work expression,

$$W_{\text{net}} = C_p(T_1 - T_4) = 0.$$

Thus, at maximum thermal efficiency, the net power output is zero.

1.8 Problem 6: Net Power Output and Required Molar Flow Rate

The net work is

$$W_{\text{net}} = C_p [(T_3 - T_4) - (T_2 - T_1)].$$

Substituting numerical values,

$$W_{\text{net}} = 29.1 \text{ J mol}^{-1}\text{K}^{-1} [(1400 - 648) - (648 - 300)] = 11756.4 \text{ J/mol}.$$

To produce 50 MW of power,

$$\dot{W} = \dot{n}W_{\text{net}}, \quad \dot{n} = \frac{\dot{W}}{W_{\text{net}}}.$$

Thus,

$$\dot{n} = \frac{50 \times 10^6 \text{ J/s}}{11756.4 \text{ J/mol}} = 4253 \text{ mol/s}.$$

To convert to MMSCFD, assume standard conditions of 1 atm and 60°F. Using

$$1 \text{ mol} = 0.836 \text{ scf},$$

the volumetric flow rate is

$$\dot{V} = (4253)(0.836) = 3555.5 \text{ scf/s}.$$

Converting to standard cubic feet per day,

$$(3555.5)(86400) = 307195200 \text{ scf/day}.$$

Thus,

$$307195200 \text{ scf/day} = 307.2 \text{ MMSCFD}.$$

1.9 Excel Implementation for Problem A

Excel screenshot for Problem A below.

	A	B	C	D	E
1	Data				
2	Heat Capacity, J/(mol K)	29.1			
3	Heat Capacity Ratio	1.4			
4					
5					
6					
7		Compressor Inlet	Compressor Outlet	Combustor Outlet	Turbine Outlet
8	Temperature, C	26.85	374.92	1126.85	374.92
9	Pressure, Bar	1	14.82	14.82	1
10	Molar Flow Rate, kmol/s	1	1	1	1
11					
12					
13	Maximum Net Power, kW	11756 kW			
14	Efficiency at Maximum Net Power	53.70%			
15					
16	Maximum Efficiency	78.57%			
17	Net Power at Maximum Efficiency, kW	0kW			
18					
19	Molar Flow Rate for 50MW Net Power, MMSCFD	307.2 MMSCFD			

Figure 2: Excel implementation of Brayton cycle calculations for Problem A.

2 Problem B: Vapor–Liquid Equilibrium Calculation

2.1 Background

Vapor–liquid equilibrium (VLE) calculations are central to the analysis and design of separation processes such as flash drums, distillation columns, and phase separators in chemical engineering. For multicomponent hydrocarbon mixtures, the vapor fraction is commonly determined using the Rachford–Rice equation, which enforces overall material balance consistency between vapor and liquid phases through component equilibrium ratios (K-values). Because this equation is nonlinear in the vapor fraction, analytical solutions are generally not possible, making numerical root-finding methods necessary. Techniques such as the Newton and Secant methods are widely used due to their efficiency and ease of implementation, particularly in spreadsheet environments like Excel. Understanding the mathematical structure of the vapor fraction equation, its derivative, and the convergence behavior of numerical methods is essential for reliably solving phase equilibrium problems and validating computational results in practical engineering applications.

2.2 Analytical Derivation of $\frac{dF}{d\alpha}$

The vapor fraction equation is

$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i(K_i - 1)}{1 + \alpha(K_i - 1)}.$$

2.2.1 Differentiation of a Single Term

$$f_i(\alpha) = z_i(K_i - 1) [1 + \alpha(K_i - 1)]^{-1}.$$

$$\frac{df_i}{d\alpha} = -\frac{z_i(K_i - 1)^2}{[1 + \alpha(K_i - 1)]^2}.$$

2.2.2 Differentiation of the Summation

$$\boxed{\frac{dF}{d\alpha} = -\sum_{i=1}^{14} \frac{z_i(K_i - 1)^2}{[1 + \alpha(K_i - 1)]^2}}$$

2.3 Newton Method Implementation

Excel screenshot for Newton method below.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Component	zi	Ki	Ki-1	denom	num F	F(alpha)	num Fprime	denom squared	Fprime(alpha)		alpha
2	1	0.117598	49.905777	48.90578	6.647297	5.751222	0.865197	281.267959	44.18655458	-6.365464833		0.115473
3	2	0.002775	13.445753	12.44575	2.437148	0.034537	0.014171	0.42983853	5.9396925	-0.072367135		0.115473
4	3	0.003435	3.474914	2.474914	1.285786	0.008501	0.006612	0.02104006	1.65324498	-0.012726523		
5	4	0.003171	1.282547	0.282547	1.032627	0.000896	0.000868	0.00025315	1.066317591	-0.000237406		
6	5	0.001189	0.628198	-0.3718	0.957067	-0.00044	-0.00046	0.00016436	0.915977066	-0.000179441		
7	6	0.000793	0.477677	-0.52232	0.939686	-0.00041	-0.00044	0.00021635	0.883009396	-0.000245011		
8	7	0.033971	0.064418	-0.93558	0.891966	-0.03178	-0.03563	0.02973528	0.795602524	-0.037374543		
9	8	0.044667	0.055577	-0.94442	0.890945	-0.04218	-0.04735	0.03984005	0.793782357	-0.050190145		
10	9	0.002572	0.226865	-0.77314	0.910724	-0.00199	-0.00218	0.00153738	0.829417807	-0.001853567		
11	10	0.065649	0.183574	-0.81643	0.905725	-0.0536	-0.05918	0.04375843	0.820337487	-0.05334199		
12	11	0.412276	0.092707	-0.90729	0.895232	-0.37406	-0.41783	0.3393776	0.801440612	-0.423459449		
13	12	0.032581	0.136715	-0.86329	0.900314	-0.02813	-0.03124	0.02428135	0.810565103	-0.029956074		
14	13	0.225623	0.072252	-0.92775	0.89287	-0.20932	-0.23444	0.19419741	0.797217114	-0.243594125		
15	14	0.0537	0.038234	-0.96177	0.888942	-0.05165	-0.0581	0.04967217	0.79021787	-0.062858828		
16							-2.2E-06			-7.35384907		

Figure 3: Excel implementation of the Newton method for Problem B.

2.4 Secant Method Implementation

Excel screenshot for Secant method below.

	A	B	C	D	E	F	G	H	I
1	Component	zi	Ki	Ki-1		Iteration #	alphan	F(alphan)	alpha(n+1)
2	1	0.117598	49.90578	48.90578		0	0.5	-1.22329	
3	2	0.002775	13.44575	12.44575		1	0.55	-1.37745	0.103249
4	3	0.003435	3.474914	2.474914		2	0.103249	0.097507	0.132783
5	4	0.003171	1.282547	0.282547		3	0.132783	-0.11502	0.116799
6	5	0.001189	0.628198	-0.3718		4	0.116799	-0.00967	0.115331
7	6	0.000793	0.477677	-0.52232		5	0.115331	0.00104	0.115474
8	7	0.033971	0.064418	-0.93558		6	0.115474	-8.7E-06	0.115473
9	8	0.044667	0.055577	-0.94442		7	0.115473	-7.7E-09	0.115473
10	9	0.002572	0.226865	-0.77314		8	0.115473	5.73E-14	0.115473
11	10	0.065649	0.183574	-0.81643					
12	11	0.412276	0.092707	-0.90729					
13	12	0.032581	0.136715	-0.86329					
14	13	0.225623	0.072252	-0.92775					
15	14	0.0537	0.038234	-0.96177					

Figure 4: Excel implementation of the Secant method for Problem B.

3 Conclusion

This assignment successfully combined analytical thermodynamics with numerical methods to evaluate both cycle performance and nonlinear equation solving in realistic chemical engineering contexts. For Problem A, an analytical optimization of the ideal Brayton cycle demonstrated that the compressor pressure ratio that maximizes net power output depends only on the temperature limits and the specific heat ratio. By differentiating the net work expression and applying isentropic relations, the optimal pressure ratio was found to be

$$r_p \approx 14.82,$$

a result confirmed independently using Excel Solver. At this operating condition, the compressor discharge pressure was 1482 kPa, the compressor exit temperature was approximately 648 K, and the turbine exhaust temperature matched this value, illustrating a key characteristic of Brayton cycles operating at maximum net power.

Using the optimal pressure ratio, the Brayton cycle thermal efficiency was calculated to be 53.7%, which, while substantial, was significantly lower than the corresponding Carnot efficiency of 78.6% operating between the same temperature limits. This comparison highlights the inherent irreversibilities and practical limitations of real thermodynamic cycles compared to idealized reversible ones. Additionally, the analysis showed that operating at maximum thermal efficiency for an ideal Brayton cycle leads to zero net power output, emphasizing the fundamental trade-off between efficiency and power production. This distinction is critical in engineering design, where maximizing useful power rather than efficiency alone is often the primary objective.

In Problem B, the vapor fraction of a multicomponent hydrocarbon mixture was determined by numerically solving the Rachford–Rice equation. An analytical expression for the derivative was derived and implemented in Excel to apply the Newton method, while the Secant method provided an alternative derivative-free approach. Both methods converged rapidly to a vapor fraction of

$$\alpha \approx 0.11547,$$

with results verified using Excel’s Goal Seek function. The Excel screenshots confirm stable convergence and correct implementation of the summation and derivative terms. Overall, the assignment reinforced the importance of combining analytical derivations with numerical tools and spreadsheet-based computation to solve complex engineering problems accurately and efficiently.

4 Tools Used

This assignment was completed with the help of ChatGPT